S M REDWAN KABIR

GIS Data Engineer | Spatial Data Analyst | Asset Management Professional

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PROFESSIONAL SUMMARY

GIS Specialist with 6 years of proven expertise in enterprise geospatial data management, spatial analysis, Python/SQL automation, and land use/transportation planning. Proficient in ArcGIS Pro/Enterprise/Online, delivering data-driven maps, dashboards, and reports for government agencies and private organization.

WORK EXPERIENCES

❖ **GIS Technician; Missouri Department of Conservation; Jefferson City, MO** ***July 2024 - Present***

- Asset tracking: Used VueWorks to centralize work orders, reducing maintenance costs via lifecycle visibility.
- Data quality: Conducted QA/QC in ArcGIS Pro/Enterprise, standardizing statewide asset data for reliable data management.
- SQL analytics: Queried geodatabases to deliver real-time reports, enabling data-driven decisions.
- Flood analysis: Mapped impacted infrastructure using ArcGIS Pro, calculating replacement costs for mitigation planning.
- Field collection: Deployed Field Maps/Survey123/Trimble R2 GNSS, integrating data into enterprise systems.
- Automation: Scripted Python/ModelBuilder workflows, boosting productivity.
- Web mapping: Created static/web maps via Dashboards/Experience Builder for internal/external stakeholder.
- Documentation: Wrote SOPs, improving team consistency and onboarding

❖ **Graduate Assistant; Western Michigan University; Kalamazoo, MI** ***August 2022 – April 2024***
Chicago Urban Heat Island GIS/ML Project

- Data pipeline: Automated climate data extraction and processing with Python/ArcPy APIs, enabling reproducible analysis.
- Remote sensing: Analyzed LST spatiotemporal patterns, mapping heat mitigation.
- Visuals: Designed ESRI maps/infographics for presentations.
- Validation: Applied quantile mapping, enhancing satellite data accuracy.
- ML analysis: Modeled heat-socioeconomic correlations at census tract level
- Policy Planning: Prepared policy for heat mitigation strategies to promote green urban infrastructure
- Research Deliverables: Presented research work in national and international conferences

❖ **Urban Planner, Sheltech Consultants Pvt. Ltd.; Bangladesh** ***Dec 2017 - June 2022***
Dhaka Master Plan (1,528 km²) Prepartion Project (Project Area-1500 km^2)

- Plan layouts: Created GIS frameworks for area/intervention plans via governmental process research.
- Reporting: Mapped baseline/progress data, improving transparency.
- Review: Assessed technical documentations proposals for transport/zoning impacts.
- Tools (S-T-A-R): Built ModelBuilder/Python tools to integrate utility infrastructure datasets, increasing productivity by 60%.
- Presentations: Delivered reports/maps/posters to officials/stakeholders for plan promotion and approval

EDUCATION

Master of Science (MS) in Geography — ***Spring 2024***
Western Michigan University; Kalamazoo MI; USA — *CGPA: 3.94 out of 4.00*

Bachelor of Urban and Regional Planning (BURP) — ***Spring 2017***
Khulna University of Engineering & Technology, Khulna, Bangladesh — *CGPA: 3.62 out of 4.00*

TECHNICAL SKILLS

GIS Products	:	ArcGIS Pro/Enterprise/Online, ModelBuilder, Survey123, Dashboards, Experience Builder, Spatial Analyst
Programming	:	Python (ArcPy, geopandas), R, SQL, statistical/ML analysis
Data Management	:	Geodatabases, Excel (pivots), Tableau/Power BI, asset management (VueWorks)
Environmental Planning:	:	Landsat, MODIS, Sentinel, Google Earth Engine, urban heat analysis
Public Engagement	:	Presentation design, grant proposal writing, stakeholder engagement
Other	:	Trimble GNSS, Google Earth Engine, AutoCAD, remote sensing (Landsat/MODIS)

ACCOMPLISHED PROJECTS

Data Mining and Environmental Modelling

- Data Science to Analyze Weather Trends with Python and NOAA API; View
- Machine Learning Model for LST Data; View
- Automating Landsat Image Information Extraction with Python and Earth Engine API; View
- Quantile Mapping Adjustment for Landsat LST from MODIS: View
- LST Analysis using Geopandas; View

Interactive Web Maps and Dashboards

- Real-time 911 public safety call dashboard creation using ArcGIS dashboard for Redland County, CA View
- Interactive Flood Monitoring Dashboard and Online Map for Missouri Dept of Conservation View
- Using geopandas to extract different layers of dataset for Forest Fire in California View
- Geocoding and drive time analysis of Michigan clinic data to the census tract data View

Transportation Projects

- Conducted drive-time analysis and accessibility modeling for healthcare and transportation equity in Michigan
- Developed a real-time transportation data dashboard integrating public safety datasets for local government use
- Applied kernel density and hotspot analysis to evaluate pedestrian and transit incident data for safety planning
- Developed GIS models and Python scripts to analyze statewide flood-affected conservation infrastructure, providing valuable risk assessment data for mitigation strategies

CERTIFICATIONS

ESRI Certificates View

- Going Places with Spatial Analysis; Instructor Led Training (6 Weeks) — *March 2025*
- ArcGIS Online: Essential Workflows 1 Day (8 Hours); Instructor-Led Training — *Dec 2024*
- Make an Impact with Modern Geo Apps; Instructor Led Training (4 weeks) — *Oct 2024*
- Python for Everyone; ESRI Web Course — *Jan 2023*
- Creating Origin-Destination Cost Matrix in ArcGIS; ESRI Web Course — *Nov 2022*
- Crime Analysis Solution: Data Management; ESRI Web Course — *Nov 2022*